Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IX

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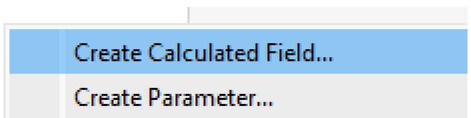
Class:IT	Batch:1
Date of Assignment:01-09-2026	Date/Time of Submission:01-09-2026

A: Identify and Handle Missing or Inconsistent Data
Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)



Physics_Clean

```
FLOAT(REPLACE(STR([Physics]), '%', ''))
```

Attendance_Clean

```
FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))
```

Step 3: Handle Missing (Null) Numeric Values via Imputation

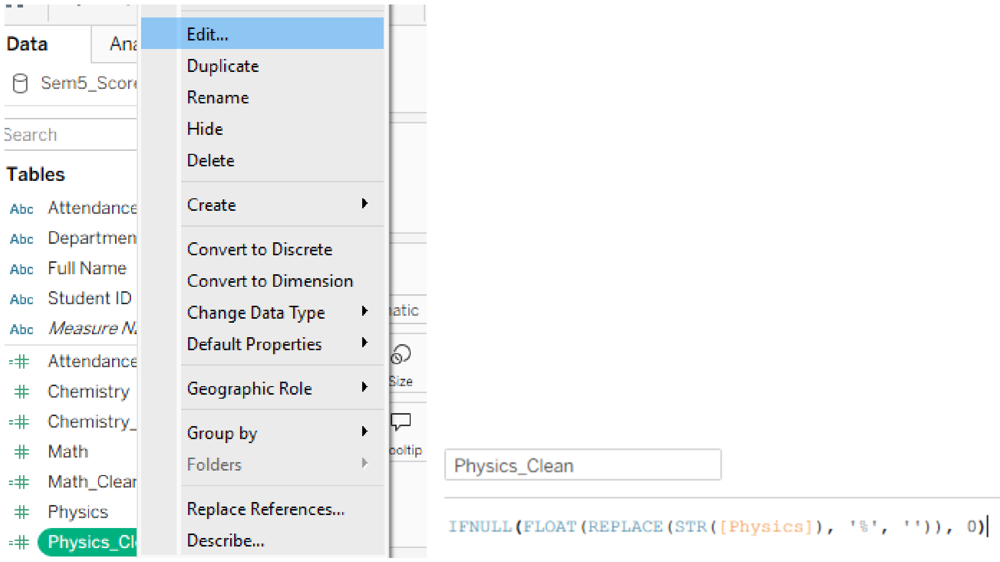
Math_Clean

```
IFNULL([Math], 0)
```

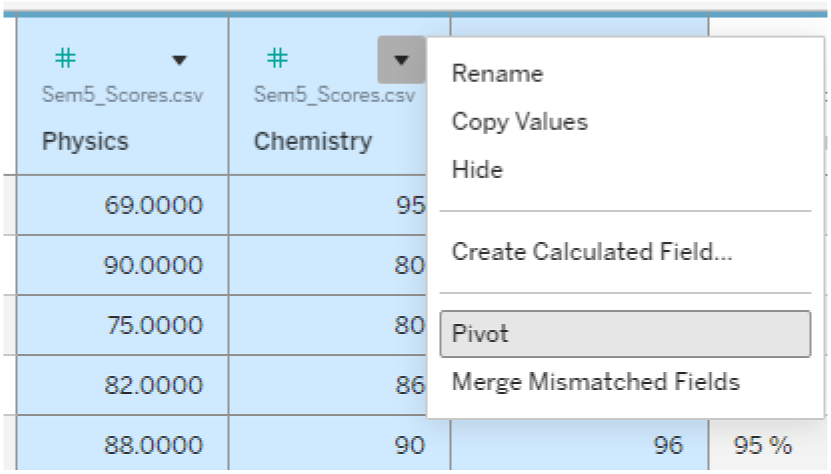
Chemistry_Clean

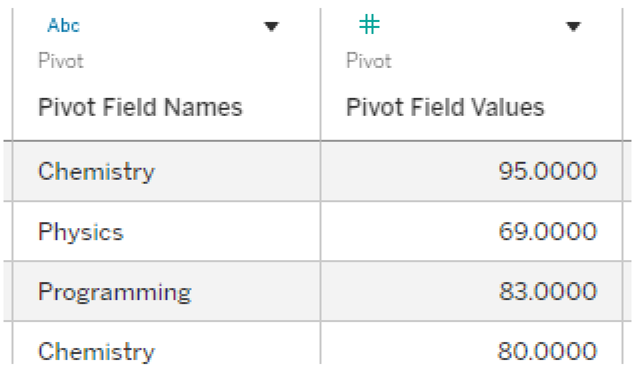
```
IFNULL([Chemistry], 0)
```

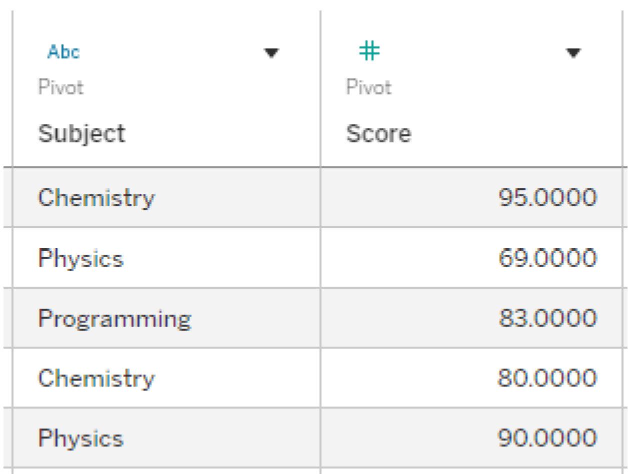
Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

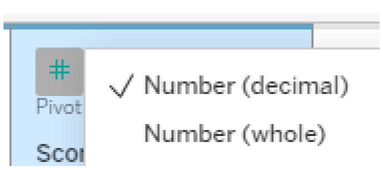


B: Convert Wide Data into Tall Format

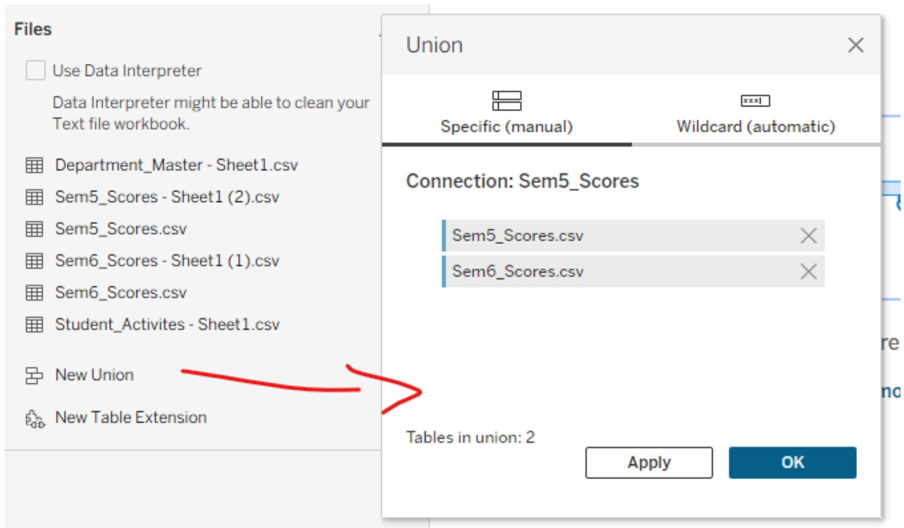








C: Perform Union of Multiple Files



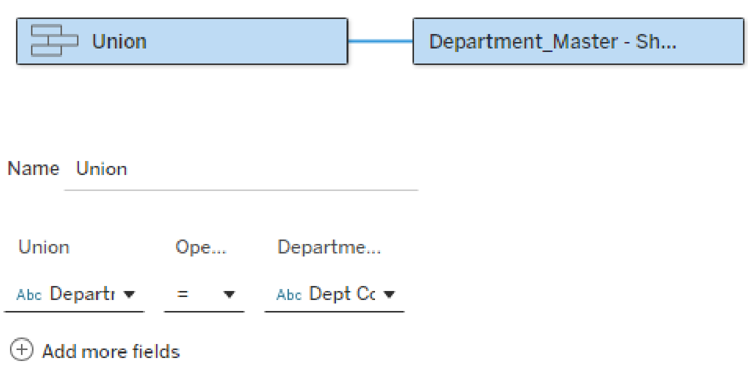
Step 3: Verify the Merged Data Grid

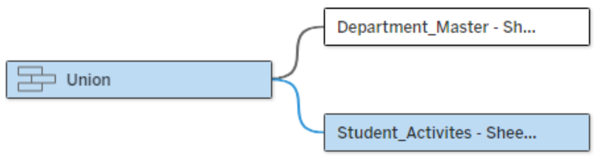
Step 4: Create a Clean Semester Identifier Dimension

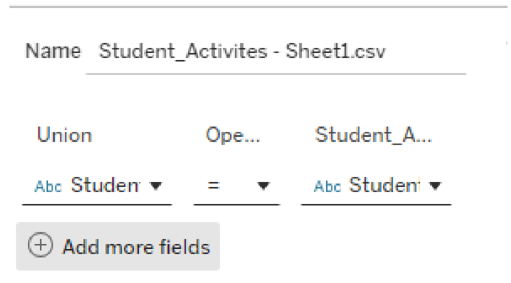
Semester

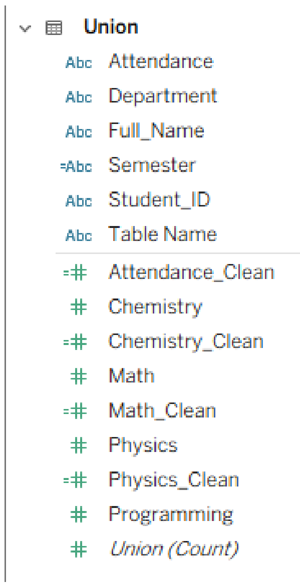
```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

D: Execute Cross-Database Joins

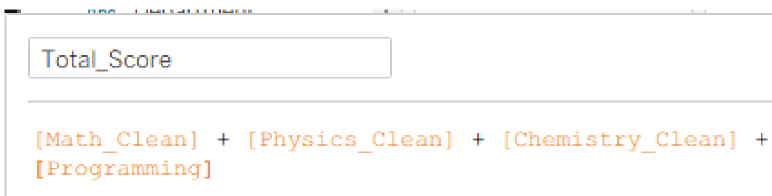


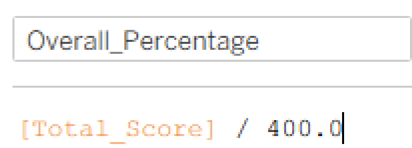


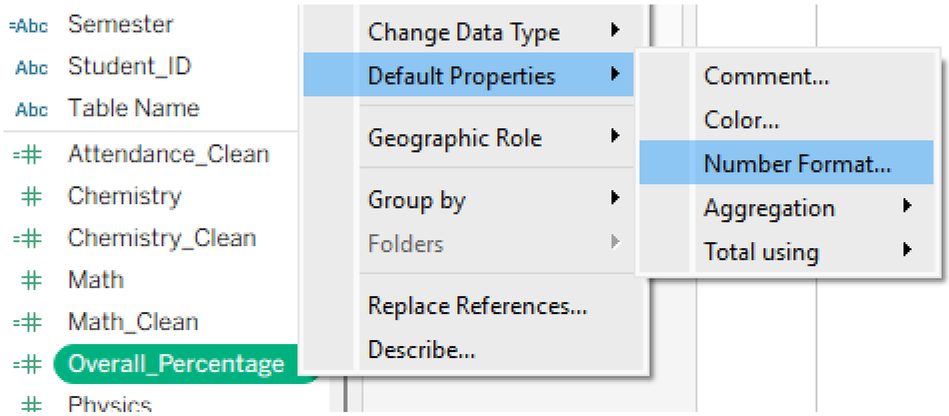


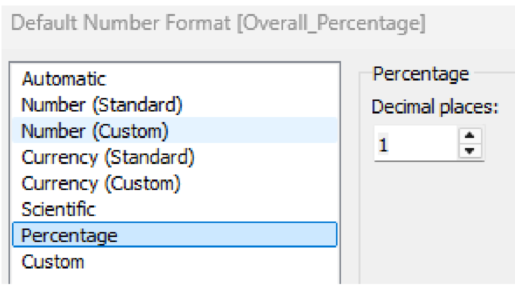


E: Restructure Data for Better Visualization







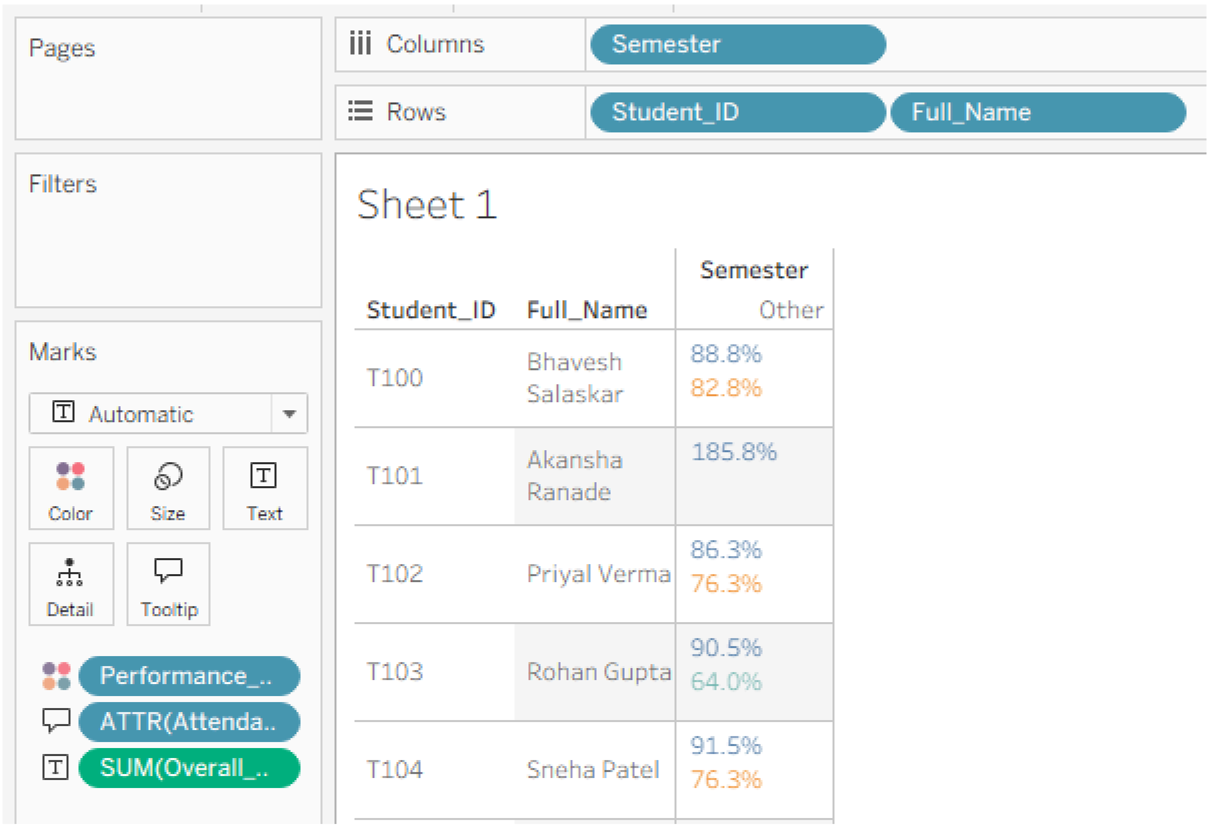


Performance_Band

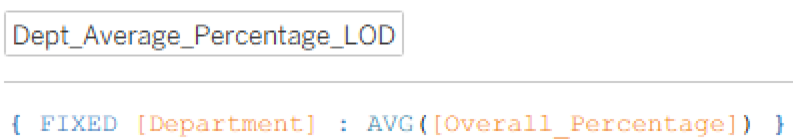
```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END
```

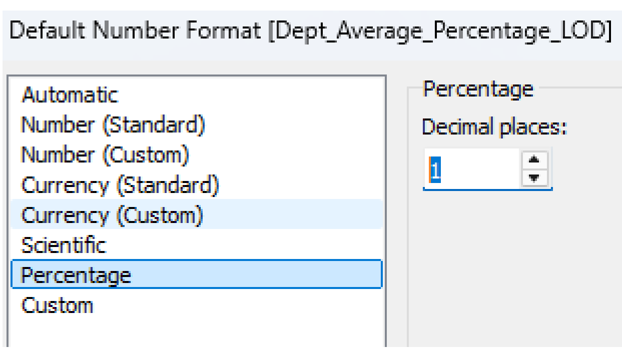
Attendance_Status

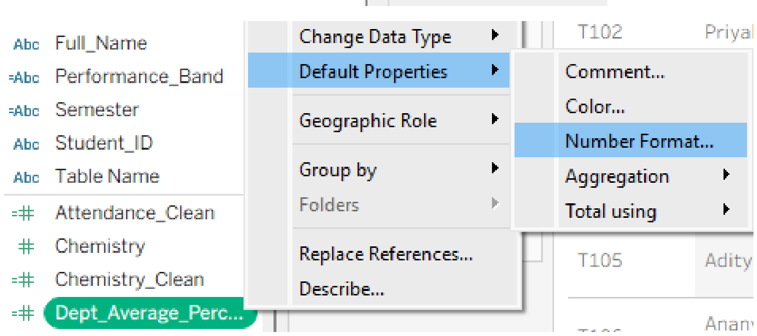
```
IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END
```



F: Handle Different Levels of Detail in Datasets







tudent_Overall_Career_Average

```
{ FIXED [Student_ID] : AVG([Overall_Percentage]) }
```

3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

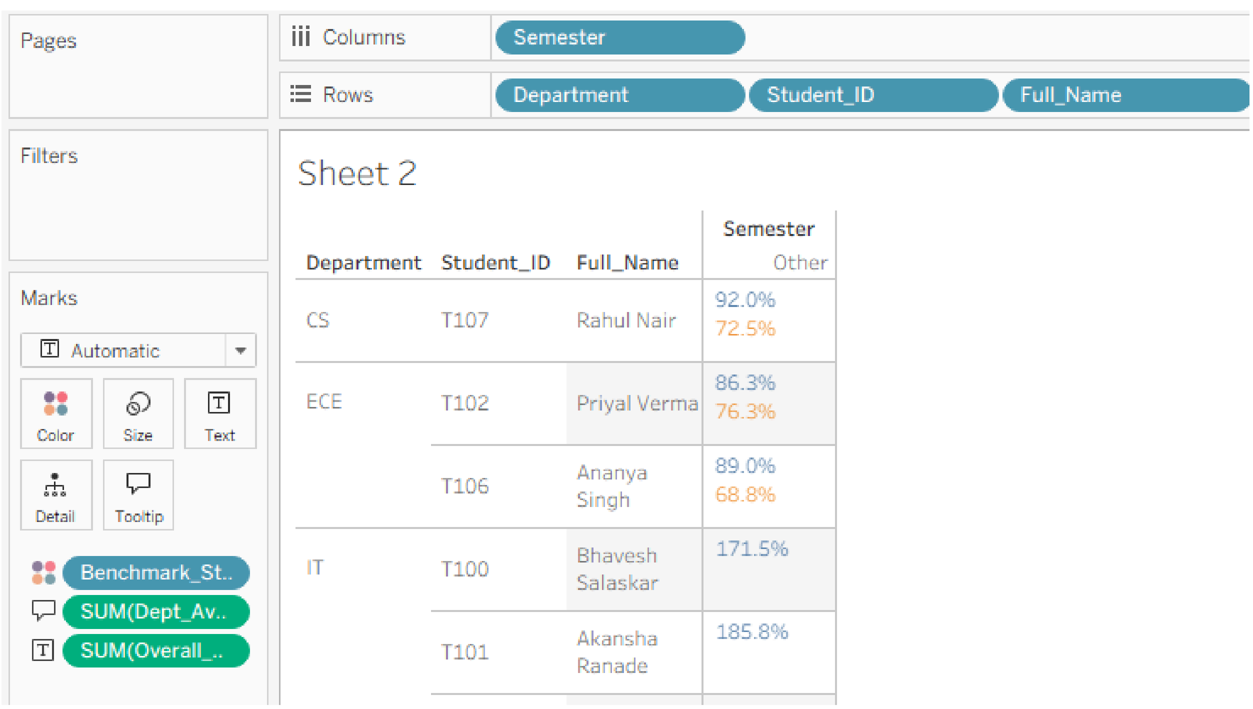
Variance_From_Dept_Average

```
[Overall_Percentage] - [Dept_Average_Percentage_LOD]
```

3. Set its Number Format to Percentage (1 decimal place).

Benchmark_Status

```
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department
Benchmark"
ELSE "Below Department Benchmark"
END
```



G: Apply Advanced Data Cleaning Techniques

Full_Name_Clean

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -
1)) +
" " +
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1,
1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2,
LEN(TRIM([Full_Name]))))
```

First_Name

```
TRIM(SPLIT([Full_Name], " ", 1))
```

Last_Name

```
TRIM(SPLIT([Full_Name], " ", -1))
```

Dept_Branch

```
SPLIT([Department], "-", 1)
```

Section_ID

```
SPLIT([Department], "-", 2)
```

Student_Display_Label

```
[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"
```